



CLASSROOM CONTACT PROGRAMME

(Academic Session : 2022 - 2023)

JEE(Advanced)
PART TEST
05-11-2022

JEE(Main + Advanced) : ENTHUSIAST COURSE [PHASE : TEAS, TOAS, TAAS & TNAS] (SCORE-I)

PAPER-1

Time : 3 Hours

Maximum Marks : 180

READ THE INSTRUCTIONS CAREFULLY

GENERAL :

1. This sealed booklet is your Question Paper. Do not break the seal till you are told to do so.
2. Use the Optical Response sheet (ORS) provided separately for answering the questions.
3. Blank spaces are provided within this booklet for rough work.
4. Write your name, form number and sign in the space provided on the back cover of this booklet.
5. After breaking the seal of the booklet, verify that the booklet contains **32** pages and that all the **18** questions in each subject and along with the options are legible. If not, contact the invigilator for replacement of the booklet.
6. You are allowed to take away the Question Paper at the end of the examination.

OPTICAL RESPONSE SHEET :

7. The ORS will be collected by the invigilator at the end of the examination.
8. Do not tamper with or mutilate the ORS. **Do not use the ORS for rough work.**
9. Write your name, form number and sign with pen in the space provided for this purpose on the ORS. **Do not write any of these details anywhere else on the ORS.** Darken the appropriate bubble under each digit of your form number.

DARKENING THE BUBBLES ON THE ORS :

10. Use a **BLACK BALL POINT PEN** to darken the bubbles on the ORS.
11. Darken the bubble  **COMPLETELY.**
12. The correct way of darkening a bubble is as : 
13. The ORS is machine-gradable. Ensure that the bubbles are darkened in the correct way.
14. Darken the bubbles **ONLY IF** you are sure of the answer. There is **NO WAY** to erase or "un-darken" a darkened bubble.
15. Take **$g = 10 \text{ m/s}^2$** unless otherwise stated.

QUESTION PAPER FORMAT :

16. The question paper has three parts : Physics, Chemistry and Mathematics.

Please see the last page of this booklet for rest of the instructions

DO NOT BREAK THE SEALS WITHOUT BEING INSTRUCTED TO DO SO BY THE INVIGILATOR

SOME USEFUL CONSTANTS

Atomic No. : H = 1, B = 5, C = 6, N = 7, O = 8, F = 9, Al = 13, P = 15, S = 16, Cl = 17, Br = 35, Xe = 54, Ce = 58

Atomic masses : H = 1, Li = 7, B = 11, C = 12, N = 14, O = 16, F = 19, Na = 23, Mg = 24, Al = 27, P = 31, S = 32, Cl = 35.5, Ca = 40, Fe = 56, Br = 80, I = 127, Xe = 131, Ba = 137, Ce = 140,

- | | |
|------------------------------------|--|
| • Boltzmann constant | $k = 1.38 \times 10^{-23} \text{ JK}^{-1}$ |
| • Coulomb's law constant | $\frac{1}{4\pi\epsilon_0} = 9 \times 10^9$ |
| • Universal gravitational constant | $G = 6.67259 \times 10^{-11} \text{ N-m}^2 \text{ kg}^{-2}$ |
| • Speed of light in vacuum | $c = 3 \times 10^8 \text{ ms}^{-1}$ |
| • Stefan-Boltzmann constant | $\sigma = 5.67 \times 10^{-8} \text{ Wm}^{-2}\text{-K}^{-4}$ |
| • Wien's displacement law constant | $b = 2.89 \times 10^{-3} \text{ m-K}$ |
| • Permeability of vacuum | $\mu_0 = 4\pi \times 10^{-7} \text{ NA}^{-2}$ |
| • Permittivity of vacuum | $\epsilon_0 = \frac{1}{\mu_0 c^2}$ |
| • Planck constant | $h = 6.63 \times 10^{-34} \text{ J-s}$ |

Space for Rough Work

HAVE CONTROL → HAVE PATIENCE → HAVE CONFIDENCE ⇒ 100% SUCCESS

BEWARE OF NEGATIVE MARKING

PART-1 : PHYSICS

SECTION-I (i) : (Maximum Marks: 24)

- This section contains **SIX (06)** questions.
- Each question has **FOUR** options. **ONE OR MORE THAN ONE** of these four option(s) is (are) correct answer(s).
- For each question, choose the option(s) corresponding to (all) the correct answer(s)
- Answer to each question will be evaluated according to the following marking scheme:

Full Marks : +4 If only (all) the correct option(s) is (are) chosen.

Partial Marks : +3 If all the four options are correct but **ONLY** three options are chosen.

Partial Marks : +2 If three or more options are correct but **ONLY** two options are chosen and both of which are correct.

Partial Marks : +1 If two or more options are correct but **ONLY** one option is chosen and it is a correct option.

Zero Marks : 0 If none of the options is chosen (i.e. the question is unanswered).

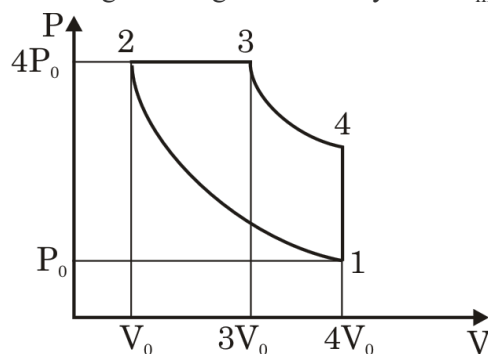
Negative Marks : -2 In all other cases.

- **For Example :** If first, third and fourth are the **ONLY** three correct options for a question with second option being an incorrect option; selecting only all the three correct options will result in +4 marks. Selecting only two of the three correct options (e.g. the first and fourth options), without selecting any incorrect option (second option in this case), will result in +2 marks. Selecting only one of the three correct options (either first or third or fourth option), without selecting any incorrect option (second option in this case), will result in +1 marks. Selecting any incorrect option(s) (second option in this case), with or without selection of any correct option(s) will result in -2 marks.

1. An element X is brought to an ionized hydrogenic ground state. It is then converted to a fully ionized form by absorption of monochromatic light of wavelength 1.42 nm ($1 \text{ nm} = 10^{-9} \text{ m}$). (An ionized hydrogenic state is an ion with only one electron orbiting the nucleus.) Which of the following statement(s) is/are true ? (Take $R = 1.1 \times 10^7 / \text{m}$)
 - (A) The atomic number of the element is 6.
 - (B) Its first Bohr orbit has a radius of 0.053 nm.
 - (C) According to the Bohr model, the angular momentum of the hydrogenic ground state of the element X is about 10^{-34} J s .
 - (D) It is possible to reconvert it to the hydrogenic ground state by absorption of an electron followed by the emission of a photon of wavelength 1.42 nm.

Space for Rough Work

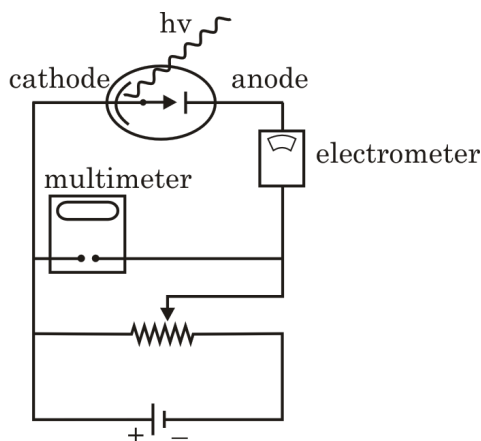
2. Let us observe the following simplified model of the Earth's atmosphere where the atmosphere layer surrounding the Earth a) reflects $\mu = 30\%$ of the Sun's radiation falling on it back to the cosmos and lets the rest of the radiation pass through it without absorbing any of it; b) absorbs completely all the radiation coming from the Earth's surface. The Solar constant is $W_0 = 1400 \text{ W/m}^2$. You can assume earth to be a black body. Assume that the thickness of atmosphere is small compared to the radius of the earth. (σ is stefan's constant)
- (A) The average energy received by the earth surface per unit area is 980 W/m^2
- (B) The average energy radiated by earth surface per unit area is 245 W/m^2
- (C) The temperature of earth surface is $T_m = \sqrt[4]{\frac{W_0 (1 - \mu)}{2\sigma}}$
- (D) The power radiated by the atmosphere in the outer space is equal to the power transmitted by it to the earth
3. There is one mole diatomic ideal gas doing cyclic process 1-2-3-4-1. The process consists of an isotherm, an isochore, a polytropic process, and an isobar. The molar heat capacity of gas during polytropic process is $C = 0.5R$. Minimal temperature of the gas during the whole cycle is $T_{\min} = 300 \text{ K}$.



- (A) The maximum temperature of the gas during the cycle is during isothermal process.
- (B) The polytropic process is given by $T \sqrt{V} = \text{constant}$
- (C) The pressure at point 4 is $3P_0/2$
- (D) The maximum temperature is 900 K .

Space for Rough Work

4.



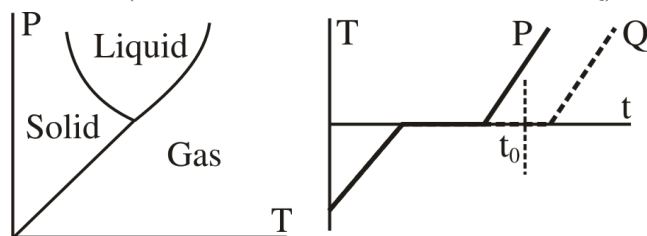
In the photoelectric effect experiment shown here, the electrometer measures current and multimeter the voltage across the photoelectric tube. The tube is illuminated by radiation from a mercury lamp. The mercury lamp radiates the wavelengths given in the chart below:

Mercury Spectrum (Hg)		
Color	Frequency ($\times 10^{14}$ Hz)	Wavelength (nm)
Red	4.34	690.8
Yellow	5.19	578.0
Green	5.49	546.1
Blue	6.88	435.8
Violet	7.41	404.7
Ultraviolet	8.20	365.5

- (A) If the work function of cathode is 1.5 eV, all the wavelengths in mercury spectrum can liberate photoelectrons from the cathode
- (B) If the resistance of the potentiometer is adjusted to be at the leftmost point, the reading in the electrometer will be maximum
- (C) If the resistance of the potentiometer is adjusted to be at the rightmost point, the reading in the electrometer will be maximum
- (D) The multimeter reads 1 V, and electrometer just reads zero. The work function of the cathode is 2.39 eV approx.

Space for Rough Work

5. A certain one mole of diatomic gas has the same molar specific heats as an ideal gas but a slightly different equation of state : $PV = R(T + \alpha T^2)$, $\alpha = 0.001 \text{ K}^{-1}$. The temperature of the gas is raised from $T_1 = 300 \text{ K}$ to T_2 at constant pressure. It is found that work done by the gas is 70% higher than what would be on an ideal gas. Choose the correct statement(s).
- (A) $T_2 = 400 \text{ K}$, internal energy increases by 250 R.
- (B) $T_2 = 400 \text{ K}$, internal energy increases by 350 R.
- (C) Total heat absorbed in the process is 450 R.
- (D) Total heat absorbed in the process is 420 R.
6. Two experiments P and Q are performed with water in various state of phases. Both experiments are performed at constant pressures. Heat is supplied to the systems at the same constant rate in both the experiments. The latent heat of fusion, vaporization and sublimation of water are respectively 334 kJ/kg, 2501 kJ/kg and 2834 kJ/kg. The phase diagram of water and the temperature-time schematic plots of the two experiments are depicted below (bold curve for P and dashed curve for Q). Then one can conclude that



- (A) the initial mass of water in P is equal to that in Q.
- (B) at time t_0 , P is in a pure phase while Q is in a mixed phase.
- (C) experiment P has been performed at a pressure higher than Q.
- (D) at $t > t_0$, Q goes to the liquid state.

Space for Rough Work

SECTION-I (ii) : (Maximum Marks: 12)

- This section contains **FOUR (04)** questions.
- Each **question has matching lists**. The codes for the lists have choices (A), (B), (C) and (D) out of which **ONLY ONE is correct**
- For each question, marks will be awarded in one of the following categories :

Full Marks : +3 If ONLY the correct option is chosen.

Zero Marks : 0 If none of the options is chosen (i.e. the question is unanswered)

Negative Marks : -1 In all other cases

7. Match the List-I from List-II

List-I		List-II	
(I)	strong nuclear force	(P)	acts on all elementary particles
(II)	weak nuclear force	(Q)	responsible for binding of nucleons
(III)	gravitational force	(R)	causes beta decay
(IV)	electromagnetic force	(S)	tension
		(T)	weight

- (A) I → P,Q;II → P,Q,R;III → T;IV → S
- (B) I → Q;II → P,R;III → P,T;IV → S
- (C) I → Q;II → R;III → S,T;IV → Q,S
- (D) I → Q,R;II → P,R;III → P,S,T;IV → S

Space for Rough Work

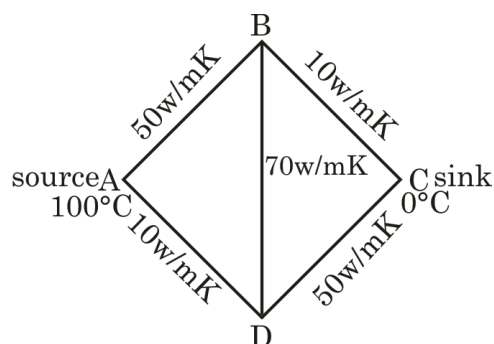
8. Consider an electron subjected to forces as shown in list-I. List-II gives options about its de-Broglie wavelength. Match them

List-I		List-II	
(I)	Electron moving under the influence of a uniform magnetic field	(P)	λ increases/may increase
(II)	Electron moving in a photoelectric tube where collector is at a positive potential w.r.t. emitter	(Q)	λ decreases/may decrease
(III)	Electron moving in a photoelectric tube where collector is at a negative potential w.r.t. emitter	(R)	λ remains constant
(IV)	Electron in ground state of H-atom absorbs a photon and moves to higher excited state and returns back.	(S)	λ may increase and then decrease
		(T)	λ may decrease & then increase.

- (A) I $\rightarrow$ P; II $\rightarrow$ P; III $\rightarrow$ S; IV $\rightarrow$ T
 (B) I $\rightarrow$ R; II $\rightarrow$ Q; III $\rightarrow$ P; IV $\rightarrow$ S
 (C) I $\rightarrow$ R; II $\rightarrow$ Q; III $\rightarrow$ P, S; IV $\rightarrow$ S
 (D) I $\rightarrow$ R; II $\rightarrow$ P; III $\rightarrow$ P; IV $\rightarrow$ T

Space for Rough Work

9. Consider a set of 5 rods of identical dimensions ($\ell = 10 \text{ cm}$, $A = 1 \text{ mm}^2$) and coefficient of thermal conductivity as shown. Find the rate of heat flow in different rods & rate of heat flow to the sink.

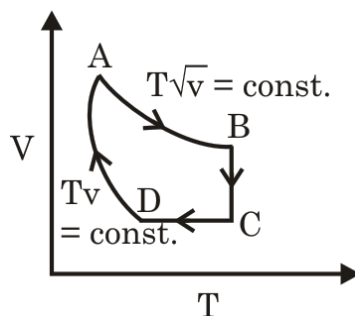


	List-I		List-II
(I)	From source	(P)	20 mw
(II)	AB	(Q)	6 mw
(III)	AD	(R)	14 mw
(IV)	BD	(S)	26 mw
		(T)	12 mw

- (A) I $\rightarrow$ S; II $\rightarrow$ R; III $\rightarrow$ Q; IV $\rightarrow$ T
 (B) I $\rightarrow$ S; II $\rightarrow$ R; III $\rightarrow$ T; IV $\rightarrow$ Q
 (C) I $\rightarrow$ S; II $\rightarrow$ T; III $\rightarrow$ Q; IV $\rightarrow$ P
 (D) I $\rightarrow$ S; II $\rightarrow$ P; III $\rightarrow$ Q; IV $\rightarrow$ R

Space for Rough Work

10. In the VT diagram shown, we have 4 processes forming a thermodynamic cycle. The gas is mixture of hydrogen & helium in mole ratio 1 : 1. Comment on the entropy of gas.



List-I		List-II	
(I)	AB	(P)	Entropy at end is same as entropy at beginning
(II)	BC	(Q)	Entropy at end is less than entropy at beginning
(III)	CD	(R)	Entropy at end is more than entropy at beginning
(IV)	DA	(S)	Entropy increases and then decreases
		(T)	Entropy decreases and then increases.

- (A) I $\rightarrow$ P; II $\rightarrow$ Q; III $\rightarrow$ Q; IV $\rightarrow$ R
 (B) I $\rightarrow$ Q; II $\rightarrow$ Q; III $\rightarrow$ Q; IV $\rightarrow$ R
 (C) I $\rightarrow$ S; II $\rightarrow$ T; III $\rightarrow$ Q; IV $\rightarrow$ R
 (D) I $\rightarrow$ P; II $\rightarrow$ Q; III $\rightarrow$ Q; IV $\rightarrow$ Q

Space for Rough Work

SECTION-II : (Maximum Marks: 24)

- This section contains **EIGHT (08)** questions. The answer to each question is a **NUMERICAL VALUE**.
- For each question, enter the correct numerical value of the answer in the place designated to enter the answer. If the numerical value has more than two decimal places, **truncate/round-off** the value to **Two** decimal places; e.g. 6.25, 7.00, -0.33, -.30, 30.27, -127.30, if answer is 11.36777..... then both 11.36 and 11.37 will be correct)
- Answer to each question will be evaluated according to the following marking scheme:

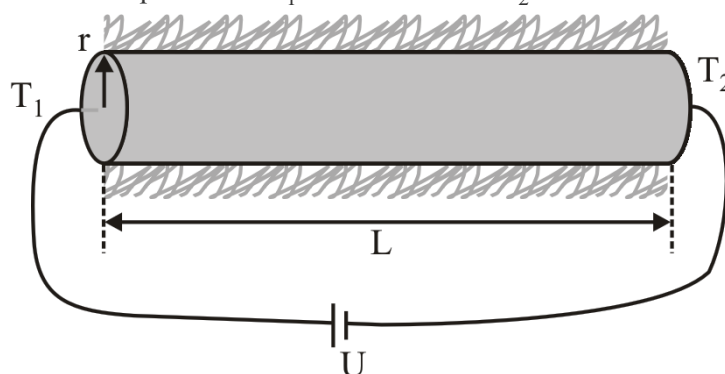
Full Marks : +3 If ONLY the correct numerical value is entered.

Zero Marks : 0 In all other cases.

1. A parallel light beam of intensity 4 W/m^2 strikes a perfectly reflecting mirror at an angle of 30° to the normal. It exerts a certain pressure on the surface of the mirror. Another light beam of intensity I strikes another plane surface at the same angle as the first case having absorptivity 0.3 and reflectivity 0.7. It is found to exert the same pressure on the surface. What is the intensity of the second beam (in W/m^2)
2. A balloon filled with helium can lift a weight with a mass up to $M = 200 \text{ kg}$. What is the volume of the balloon V (in m^3) ? The volume of the weight is negligible. The mass of the balloon's shell is included in the mass of the weight. The air density is $\rho = 1.2 \text{ kg/m}^3$, air pressure $p = 100 \text{ kPa}$, air temperature $T = 27^\circ \text{C}$. The molar mass of helium is $\mu = 4.0 \text{ g/mol}$, universal gas constant is $R = 25/3 \text{ J} \cdot \text{mol}^{-1} \cdot \text{K}^{-1}$.
3. A moving Polonium atom (^{210}Po) undergoes alpha decay such that the daughter nucleus (^{206}Pb) is at rest but alpha particle carries the energy. What is the initial kinetic energy (in keV) of the Polonium atom. Given mass of $^{210}\text{Po} = 209.98286 \text{ amu}$, $^{206}\text{Pb} = 205.97446 \text{ amu}$, $^4\text{He} = 4.00260 \text{ amu}$, $1 \text{ amu} \equiv 930 \text{ MeV}$.

Space for Rough Work

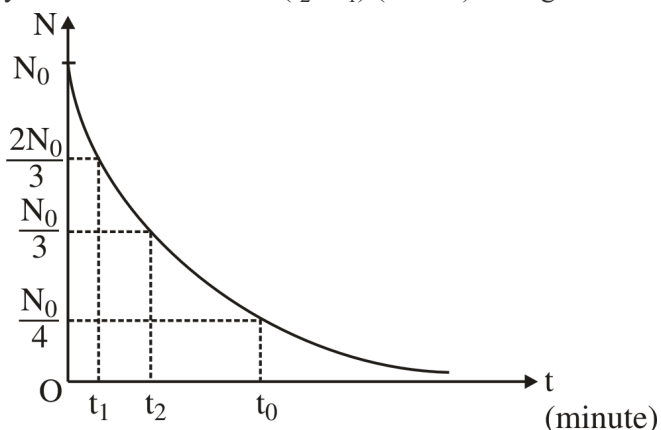
4. A neutron is fired towards a stationary hydrogen atom in ground state with a kinetic energy of 24.2 eV. The Hydrogen atom gets excited. If the collision is partially elastic, what is the maximum energy (in eV) of the emitted photon when hydrogen atom gets de-excited ?
5. Two samples of same liquid have volumes 400 cc and 220 cc and their temperature are 10°C and 110°C respectively. Find the final temperature (in °C) of the mixture if the two samples are mixed. Assume no heat exchange with the surroundings. The specific heat capacity of the liquid does not depend on the temperature. Volume coefficient of thermal expansion is $10^{-3}/^{\circ}\text{C}$.
6. A cylindrical cable with length $L = 1.0$ m and area $r = 1.0 \text{ mm}^2$ is connected to the battery with emf $U = 0.1$ V. Most of the surface of the cable is heat insulated, except two flat ends, which in steady state condition have temperatures $T_1 = 300.0$ K and $T_2 = 400.0$ K



Determine temperature T_m (in K) in the middle of the cable, if material of the cable has resistivity $\rho = 1.0 \times 10^{-6} \text{ Ohm}\cdot\text{m}$ and thermal conductivity $k = 10.0 \text{ W/mK}$.

Space for Rough Work

7. In an X-ray tube, after an electron has been removed from an inner shell of a target atom, an electron from outer shell falls into the vacancy and the excess energy is usually released in form of a photon. Many a times an atom chooses to release this excess energy by ejecting another electron. The ejected electron is called Auger electron. In one such event, the bombarding electron removed an electron from K shell of a target atom. An electron from L shell falls to occupy the vacant K shell position, and the excess energy is used by the atom to eject an Auger electron from L shell. What will be kinetic energy (in KeV) of such an Auger electron if the ionization energies for K and L shell of the atom are 62 KeV and 20 KeV, respectively.
8. The population of active nuclei in a sample of radioactive isotope has been plotted with time in figure. Look at the graph carefully and find the value of $(t_2 - t_1)$ (in min). It is given that $t_0 = 50$ minute.



Space for Rough Work

PART-2 : CHEMISTRY
SECTION-I (i) : (Maximum Marks: 24)

- This section contains **SIX (06)** questions.
- Each question has **FOUR** options. **ONE OR MORE THAN ONE** of these four option(s) is (are) correct answer(s).
- For each question, choose the option(s) corresponding to (all) the correct answer(s)
- Answer to each question will be evaluated according to the following marking scheme:

Full Marks : +4 If only (all) the correct option(s) is (are) chosen.

Partial Marks : +3 If all the four options are correct but **ONLY** three options are chosen.

Partial Marks : +2 If three or more options are correct but **ONLY** two options are chosen and both of which are correct.

Partial Marks : +1 If two or more options are correct but **ONLY** one option is chosen and it is a correct option.

Zero Marks : 0 If none of the options is chosen (i.e. the question is unanswered).

Negative Marks : -2 In all other cases.

- **For Example :** If first, third and fourth are the **ONLY** three correct options for a question with second option being an incorrect option; selecting only all the three correct options will result in +4 marks. Selecting only two of the three correct options (e.g. the first and fourth options), without selecting any incorrect option (second option in this case), will result in +2 marks. Selecting only one of the three correct options (either first or third or fourth option), without selecting any incorrect option (second option in this case), will result in +1 marks. Selecting any incorrect option(s) (second option in this case), with or without selection of any correct option(s) will result in -2 marks.

-
1. Select correct about interhalogen compound XX'_3 in which $X = I$ & $X' = Cl$
- (A) It form when I_2 react with Cl_2 in equimolar amount
- (B) It is orange solid
- (C) Molecular geometry around iodine is tetrahedral
- (D) It conduct electricity in its molten form
-

Space for Rough Work

2. Ore/mineral in which at least one metal is in +2 oxidation state.
- (A) Wolframite (B) Chromite
(C) Magnetite (D) Pyrolusite
3. Select correct about froth flotation process :-
- (A) The polar head groups of detergent coat the surface of mineral particles
(B) Non polar tails point outwards, making the detergent coated mineral particles hydrophilic
(C) Detergent coated mineral particles trapped in the froth & float to the surface of mixture
(D) Gangue particles attach to water and sinks to bottom of flotation vessel.
4. Consider reaction and select correct option(s) :-
- $$\text{MnO}_2(\text{hydrated}) + \text{SO}_2 \xrightarrow{\text{aq.}/0^\circ\text{C}} (\text{X}) \text{ Manganese (II) sulphate} + (\text{Y}) \text{ Manganese (II) dithionate}$$
- (A) It's a redox change
(B) SO_2 is oxidised by MnO_2
(C) Oxidation number of sulphur is same in anion of (X) & anion of (Y)
(D) S–O bond length is higher in (X) as compared to (Y)

Space for Rough Work

5. Cobalt(II) chloride and diethylenetriamine of empirical formula $[\text{CoCl}_2(\text{dien})]$ is not a five coordinated molecule but a salt containing octahedral homoleptic cation and tetrahedral homoleptic anion. Select correct for above complex salt.
- (A) Crystal field splitting energy of cation is higher as compared to anion
(B) Anion absorb orange light for d-d transition
(C) Cation have geometrical isomeric forms
(D) Cation is diamagnetic but anion is paramagnetic
6. In which reaction(s) ; phosphorus in reactant is trivalent but phosphorus is pentavalent in product.
- (A) Reaction of phosphorus trichloride with water at room temperature
(B) Reaction of white phosphorus with SOCl_2
(C) Reaction of white phosphorus with SO_2Cl_2
(D) Reaction of R-PCl_2 with water at room temperature (R = alkyl group)

Space for Rough Work

SECTION-I (ii) : (Maximum Marks: 12)

- This section contains **FOUR (04)** questions.
- Each **question has matching lists**. The codes for the lists have choices (A), (B), (C) and (D) out of which **ONLY ONE is correct**
- For each question, marks will be awarded in one of the following categories :

Full Marks : +3 If ONLY the correct option is chosen.

Zero Marks : 0 If none of the options is chosen (i.e. the question is unanswered)

Negative Marks : -1 In all other cases

7. Consider pair of complex/complex ion in List-I & match the correct characteristics of that pair of complex/complex ion in List-II & select most appropriate option.

List-I		List-II	
(P)	$[\text{PtBr}_2\text{Cl}_2]^{2-}$; $[\text{CoBr}_2\text{Cl}_2]^{2-}$	(1)	Same oxidation number of metal
(Q)	$[\text{Pt}(\text{NH}_3)_3\text{Cl}]\text{Br}$; $[\text{Pt}(\text{NH}_3)_3(\text{NO}_3)]\text{Cl}$	(2)	Same magnetic moment
(R)	$[\text{Cr}(\text{NH}_3)_5\text{Cl}]\text{Cl}_2$; $[\text{Cr}(\text{NH}_3)_4\text{Cl}_2]\text{Cl}$	(3)	Same shape of complex
(S)	$[\text{Co}(\text{NH}_3)_4\text{Cl}_2]^+$; $[\text{Pt}(\text{NH}_3)_2(\text{H}_2\text{O})(\text{Cl})]^+$	(4)	Same kind of isomerism
		(5)	Same number of metal-Cl linkages in co-ordination sphere

- (A) $P \rightarrow 1,3,4,5$; $Q \rightarrow 1,2,3,4$; $R \rightarrow 1,2,3$; $S \rightarrow 2,4$
- (B) $P \rightarrow 1,5$; $Q \rightarrow 1,2,3,5$; $R \rightarrow 1,2,3$; $S \rightarrow 2,4$
- (C) $P \rightarrow 1,5$; $Q \rightarrow 1,2,3,4$; $R \rightarrow 1,2,3$; $S \rightarrow 2,4$
- (D) $P \rightarrow 1,3,5$; $Q \rightarrow 1,2,3,4$; $R \rightarrow 1,2,3,4$; $S \rightarrow 1,4$

Space for Rough Work

8. Match the compounds in List-I with the descriptions in List-II and choose the correct option.

List-I		List-II	
(P)	Dinitrogen pentoxide	(1)	Prepared by heating ammonium nitrite
(Q)	Dinitrogen tetroxide	(2)	Coloured solid dissociated reversibly in two different gases
(R)	Dinitrogen monoxide	(3)	Colourless solid/liquid at -11.2°C dissociates reversibly in brown gas only at high temperature
(S)	Dinitrogen trioxide	(4)	Colourless ionic solid
		(5)	Isoelectronic with one of the ion in dinitrogen pentoxide

- (A) $P \rightarrow 4; Q \rightarrow 2; R \rightarrow 1,5; S \rightarrow 3$ (B) $P \rightarrow 2; Q \rightarrow 5; R \rightarrow 1,3; S \rightarrow 4$
 (C) $P \rightarrow 4; Q \rightarrow 3; R \rightarrow 1,5; S \rightarrow 2$ (D) $P \rightarrow 4; Q \rightarrow 3; R \rightarrow 5; S \rightarrow 2$

9. Consider the reactions of Xe given in List-I and characteristics of product in List-II.

Select most appropriate match :-

List-I		List-II	
(P)	$\text{XeF}_2 + \text{PF}_5 \rightarrow$	(1)	Xe in anionic part of product
(Q)	$\text{Xe} + \text{PtF}_6 \rightarrow$	(2)	Xe in cationic part of product
(R)	$\text{XeF}_4 + \text{SbF}_5 \rightarrow$	(3)	Octahedral anion
(S)	$\text{XeF}_6 + \text{CsF} \rightarrow$	(4)	Paramagnetic
		(5)	Only one lone pair on Xe

The correct code is :-

- (A) $P \rightarrow 2,4; Q \rightarrow 3,4; R \rightarrow 2,5; S \rightarrow 1,5$ (B) $P \rightarrow 2,3; Q \rightarrow 2,4; R \rightarrow 1,3; S \rightarrow 1,4$
 (C) $P \rightarrow 2,3,4; Q \rightarrow 2,3,4; R \rightarrow 2,3,5; S \rightarrow 1,3$ (D) $P \rightarrow 2,3; Q \rightarrow 2,3; R \rightarrow 2,3; S \rightarrow 1,5$

Space for Rough Work

10. Consider MOT for species (X_2) given in List-I and match them with correct characteristics List-II and select most appropriate answer

List-I		List-II	
(P)	B_2	(1)	No s-p mixing
(Q)	C_2	(2)	π bond order = one
(R)	O_2	(3)	No unpaired electrons
(S)	F_2	(4)	Lower bond order of X_2^- as compared to X_2
		(5)	σ bond order = zero

- (A) $P \rightarrow 3,5; Q \rightarrow 3,5; R \rightarrow 2,4,5; S \rightarrow 1,3,4$
 (B) $P \rightarrow 2,5; Q \rightarrow 3,5; R \rightarrow 1,2,4; S \rightarrow 1,3,4$
 (C) $P \rightarrow 1,2,5; Q \rightarrow 3,4,5; R \rightarrow 1,4; S \rightarrow 3,4$
 (D) $P \rightarrow 2,5; Q \rightarrow 1,3,5; R \rightarrow 1,2,4,5; S \rightarrow 1,3,4,5$

Space for Rough Work

SECTION-II : (Maximum Marks: 24)

- This section contains **EIGHT (08)** questions. The answer to each question is a **NUMERICAL VALUE**.
- For each question, enter the correct numerical value of the answer in the place designated to enter the answer. If the numerical value has more than two decimal places, **truncate/round-off** the value to **Two** decimal places; e.g. 6.25, 7.00, -0.33, -0.30, 30.27, -127.30, if answer is 11.36777..... then both 11.36 and 11.37 will be correct)
- Answer to each question will be evaluated according to the following marking scheme:

Full Marks : +3 If ONLY the correct numerical value is entered.

Zero Marks : 0 In all other cases.

1. The bad smell of water containing hydrogen sulphide can be removed by the action of chlorine (not in excess). If the hydrogen sulphide content of contaminated water is 22 (ppm) by mass, calculate the amount of Cl_2 (in grams) required to remove all the H_2S from 2×10^2 gallons of water (1 gallon = 3.785 Lt.) (Atomic weight :- H = 1, S = 32, Cl = 35.5, O = 16)

2. A 130.5 mg sample of pure pyrolusite is heated with excess conc. HCl, the resulting gas passed in to aq. solution of excess KI to produce brown solution.
Brown solution required x ml of 0.1 M $\text{Na}_2\text{S}_2\text{O}_3$ solution for completely decolorization.
Find the value of x. (Atomic weight :- H = 1, Mn = 55, O = 16, Na = 23, S = 32, K = 39, I = 127, Cl = 35.5, Zn = 65, Cu = 63.5)

3. Sphalerite $\xrightarrow{\text{Roasting}}$ solid + Gas (X)

Sylvine + conc. $\text{H}_2\text{SO}_4 \rightarrow$ other product + Gas (Y)
Gas (Y) + $\text{O}_2 \xrightarrow{\text{CuCl}_2, \Delta}$ Gas (Z) + H_2O
Gas (X) + Gas (Z) $\xrightarrow{\text{Charcoal}}$ Product gas (P)

What is the molecular weight of product gas (P).

(Atomic weight :- S = 32, O = 16, Cl = 35.5, Zn = 65, K = 39, Cu = 63.5, N = 14)

Space for Rough Work

4. Phosphorus occurs in minerals of apatite family. Write number of Ca^{2+} in the formula of fluorapatite
5. If bond length of gas phase HBr molecule is 141.5 pm and charge distribution in coulomb in gases molecule is $\text{H}^{+0.123} - \text{Br}^{-0.123}$. [Charge on one electron = $1.6 \times 10^{19} \text{C} / 4.8 \times 10^{-10} \text{esu}$]
Calculate dipole moment (in Debye) of HBr gases molecule.
6. During metallurgical extraction the sulphide ores of metals are heated with regular supply of oxygen in a furnace at a temperature below the melting of the roasted product. When 1.164 gm of certain metal sulphide was roasted in oxygen 0.972 gm of the metal oxide was formed. If oxidation number of metal is +2 in ore.
Calculate the molar mass of metal.
7. Consider bond angles values.
 $72^\circ, 90^\circ, 107^\circ, 109.5^\circ, 116^\circ, 120^\circ$
 $[\text{CoCl}_4]^{2-}, [\text{PtCl}_4]^{2-}, [\text{XeF}_5]^{-}, [\text{AlCl}_4]^{-}, [\text{PCl}_6]^{-}, \text{IF}_7, [\text{BeCl}_3]^{-}, [\text{ICl}_2]^{-}, [\text{ICl}_4]^{-}$
Which bond angle value is associated with maximum number of given species.
8. (I) 1 mole Ag + dil. $\text{HNO}_3 \rightarrow$ Metal nitrate + other products
(II) 1 mole Ag + conc. $\text{HNO}_3 \rightarrow$ Metal nitrate + other products
Find sum of the amount of nitric acid required (in gm) to completely react with given amount of silver in reaction (I) and reaction (II). (Atomic weight :- Ag = 108, H = 1, O = 16, N = 14)

Space for Rough Work

PART-3 : MATHEMATICS
SECTION-I (i) : (Maximum Marks: 24)

- This section contains **SIX (06)** questions.
- Each question has **FOUR** options. **ONE OR MORE THAN ONE** of these four option(s) is (are) correct answer(s).
- For each question, choose the option(s) corresponding to (all) the correct answer(s)
- Answer to each question will be evaluated according to the following marking scheme:

Full Marks : +4 If only (all) the correct option(s) is (are) chosen.

Partial Marks : +3 If all the four options are correct but **ONLY** three options are chosen.

Partial Marks : +2 If three or more options are correct but **ONLY** two options are chosen and both of which are correct.

Partial Marks : +1 If two or more options are correct but **ONLY** one option is chosen and it is a correct option.

Zero Marks : 0 If none of the options is chosen (i.e. the question is unanswered).

Negative Marks : -2 In all other cases.

- **For Example :** If first, third and fourth are the **ONLY** three correct options for a question with second option being an incorrect option; selecting only all the three correct options will result in +4 marks. Selecting only two of the three correct options (e.g. the first and fourth options), without selecting any incorrect option (second option in this case), will result in +2 marks. Selecting only one of the three correct options (either first or third or fourth option), without selecting any incorrect option (second option in this case), will result in +1 marks. Selecting any incorrect option(s) (second option in this case), with or without selection of any correct option(s) will result in -2 marks.

1. If
$$\begin{vmatrix} ah + bg & g & ab + ch \\ bf + ba & f & hb + bc \\ af + bc & c & bg + fc \end{vmatrix} = \lambda \begin{vmatrix} ah + bg & a & h \\ bf + ba & h & b \\ af + bc & g & f \end{vmatrix}$$
 Then λ is independent of

(A) a (B) b (C) c (D) f

Space for Rough Work

2. Let OPR be a triangle right angled at P and O is origin. OQ is angle bisector and point Q is such that ORQP is cyclic quadrilateral. If points P, Q, R are represented by complex number z_3, z_2, z_1 respectively such that $\frac{z_2^2}{z_1 z_3} = \frac{3}{2}$ then
- (A) Angles of $\triangle OPR$ are in A.P. (B) Angles of $\triangle OPR$ are $\frac{\pi}{4}, \frac{\pi}{4}, \frac{\pi}{2}$
- (C) Area of $\triangle OPR$ is $2\sqrt{2}R^2$ (D) Area of $\triangle OPR$ is $\frac{\sqrt{3}R^2}{2}$
3. Let A and B of square matrices of order 'n' such that $AB + B^2A^2 = I$ and $A^3 + B^3 = 0$ then
- (A) $BA + A^2B^2 = I$ (B) $A - B^2$ is non singular
- (C) $B - A^2$ is non singular (D) $A - B^2$ is singular
4. Let A be a non-singular square matrix of order 2 having real elements such that $|A + |A|| \operatorname{adj} A| = 0$, then which of the following is/are correct
- (A) $|A| = 1$ (B) $\operatorname{Tr}(A) = 0$
- (C) $|\operatorname{adj}(\operatorname{adj} A)| = 1$ (D) $|A - |A|| \operatorname{adj} A| = 4$
5. A line AB of length 2ℓ moves with the End A always on x-axis and the end B on line $y = 6x$, then
- (A) Locus of middle point of AB is $9x^2 + 10y^2 - 6xy = 9\ell^2$
- (B) Locus of centroid of $\triangle OAB$ is $9x^2 + 10y^2 - 6xy = 4\ell^2$
- (C) Locus of point which divides AB in ratio 2 : 1 is $(12x - 3y)^2 + (6y)^2 = 64\ell^2$
- (D) None of these
6. A straight line L passing through the point A(-2,-3) cuts the line $x + 3y = 9$ and $x + y + 1 = 0$ at B and C respectively. If $(AB)(AC) = 20$ then equation of line L can be
- (A) $x - y = 1$ (B) $x - y + 1 = 0$ (C) $3x - y + 3 = 0$ (D) $3x - y = 3$

Space for Rough Work

SECTION-I (ii) : (Maximum Marks: 12)

- This section contains **FOUR (04)** questions.
- Each **question has matching lists**. The codes for the lists have choices (A), (B), (C) and (D) out of which **ONLY ONE is correct**
- For each question, marks will be awarded in one of the following categories :

Full Marks : +3 If ONLY the correct option is chosen.

Zero Marks : 0 If none of the options is chosen (i.e. the question is unanswered)

Negative Marks : -1 In all other cases

7.

List-I		List-II	
(I)	Let ω be a non real cube root of unity then number of different elements in the set $\{(1 + \omega + \omega^2 + \dots + \omega^n)^m, m, n \in \mathbb{N}\}$	(P)	4
(II)	Let $1, \omega, \omega^2$ be cube root of unity. The least possible degree of polynomial with real coefficients having roots $2\omega, (2 + 3\omega), (2 + 3\omega^2), (2 - \omega - \omega^2)$	(Q)	5
(III)	Let $\alpha = 6 + 4i, \beta = 2 + 4i$ are two complex numbers on the complex plane. A complex number z satisfying $\text{amp} \left(\frac{z - \alpha}{z - \beta} \right) = \frac{\pi}{6}$ moves on major segment of circle whose radius is	(R)	6
(IV)	If $ z + 4 \leq 3$, z belongs to set of complex numbers the greatest value of $ z + 1 $	(S)	7
		(T)	3

(A) I $\rightarrow$ S; II $\rightarrow$ Q; III $\rightarrow$ P; IV $\rightarrow$ R

(B) I $\rightarrow$ R; II $\rightarrow$ T; III $\rightarrow$ Q; IV $\rightarrow$ S

(C) I $\rightarrow$ P; II $\rightarrow$ R; III $\rightarrow$ T; IV $\rightarrow$ Q

(D) I $\rightarrow$ Q; II $\rightarrow$ T; III $\rightarrow$ R; IV $\rightarrow$ S

Space for Rough Work

8.

List-I		List-II	
(I)	If $P = \begin{bmatrix} 1 & \alpha & 3 \\ 1 & 3 & 3 \\ 2 & 4 & 4 \end{bmatrix}$ is adjoint of 3×3 matrix A and $ A = 2$ then α is	(P)	8
(II)	If $A = \begin{bmatrix} 2 & -3 \\ -4 & 1 \end{bmatrix}$ and $B = \text{adj}(3A^2 + 12A)$ then b_{22} is divisible by where $B = [b_{ij}]_{2 \times 2}$	(Q)	5
(III)	Let A and B be two invertible matrices of order 3×3 if $\det(A B A^T) = 8$ and $\det(AB^{-1}) = 8$ then $\frac{1}{\det(BA^{-1}B^T)}$	(R)	1
(IV)	If for matrix $A = \begin{bmatrix} 1 & -\alpha \\ \alpha & \beta \end{bmatrix}$; $AA^T = I_2$, then the value of $\alpha^4 + \beta^4$ is	(S)	16
		(T)	9

(A) I $\rightarrow$ R; II $\rightarrow$ T; III $\rightarrow$ P; IV $\rightarrow$ Q

(B) I $\rightarrow$ Q; II $\rightarrow$ P, R, T; III $\rightarrow$ S; IV $\rightarrow$ R

(C) I $\rightarrow$ P; II $\rightarrow$ Q; III $\rightarrow$ T; IV $\rightarrow$ S

(D) I $\rightarrow$ S; II $\rightarrow$ T; III $\rightarrow$ Q; IV $\rightarrow$ P

Space for Rough Work

9.

List-I		List-II	
(I)	If the distance between centres of two circles of radii 3,4 units is 25 units then the length of their transverse common tangent is	(P)	24
(II)	Tangents are drawn from (4,4) to the circle $x^2 + y^2 - 2x - 2y - 7 = 0$ meet the circle at A and B. If the length of the chord AB is $3\sqrt{T}$ then value of T is	(Q)	2
(III)	Number of real solution(s) of the equation $ x - 3 ^{3x^2 - 10x + 3} = 1$	(R)	3
(IV)	$N = \sqrt{2 + \sqrt{5}} - \sqrt{6 - 3\sqrt{5}} + \sqrt{14 - 6\sqrt{5}}$	(S)	1
		(T)	4

(A) I $\rightarrow$ T; II $\rightarrow$ P; III $\rightarrow$ R; IV $\rightarrow$ S

(B) I $\rightarrow$ R; II $\rightarrow$ Q; III $\rightarrow$ T; IV $\rightarrow$ S

(C) I $\rightarrow$ P; II $\rightarrow$ Q; III $\rightarrow$ R; IV $\rightarrow$ Q

(D) I $\rightarrow$ Q; II $\rightarrow$ T; III $\rightarrow$ S; IV $\rightarrow$ P

Space for Rough Work

10.

	List-I	List-II	
(I)	$ z - 2i + z - 7i = k$ then locus of z is an ellipse of k is	(P)	1
(II)	Value of $\sum_{n=1}^5 \left(x^n + \frac{1}{x^n} \right)^2$ when $x^2 - x + 1 = 0$	(Q)	3
(III)	If imaginary part of $(z - 1)(\cos \alpha - i \sin \alpha) + (z - 1)^{-1}(\cos \alpha + i \sin \alpha)$ is zero then $ z - 1 $ is	(R)	8
(IV)	$(xy - 7)^2 = x^2 + y^2$ find possible value of xy	(S)	0
		(T)	12

(A) $I \rightarrow R, T; II \rightarrow R; III \rightarrow P; IV \rightarrow S, T$

(B) $I \rightarrow P; II \rightarrow S; III \rightarrow Q; IV \rightarrow R$

(C) $I \rightarrow Q; II \rightarrow R; III \rightarrow T; IV \rightarrow S$

(D) $I \rightarrow T; II \rightarrow Q; III \rightarrow S; IV \rightarrow P$

Space for Rough Work

SECTION-II : (Maximum Marks: 24)

- This section contains **EIGHT (08)** questions. The answer to each question is a **NUMERICAL VALUE**.
- For each question, enter the correct numerical value of the answer in the place designated to enter the answer. If the numerical value has more than two decimal places, **truncate/round-off** the value to **Two** decimal places; e.g. 6.25, 7.00, -0.33, -.30, 30.27, -127.30, if answer is 11.36777..... then both 11.36 and 11.37 will be correct)
- Answer to each question will be evaluated according to the following marking scheme:

Full Marks : +3 If ONLY the correct numerical value is entered.

Zero Marks : 0 In all other cases.

-
1. If a, b, c are respectively $p^{\text{th}}, 2q^{\text{th}}, 3r^{\text{th}}$ term of harmonic progression then the value of $\begin{vmatrix} bc & ca & ab \\ p & 2q & 3r \\ 1 & 1 & 1 \end{vmatrix}$
 2. If A, B, C are the angles of acute angle ΔABC of $\Delta = \begin{vmatrix} (\tan B + \tan C)^2 & \tan^2 A & \tan^2 A \\ \tan^2 B & (\tan A + \tan C)^2 & \tan^2 B \\ \tan^2 C & \tan^2 C & (\tan A + \tan B)^2 \end{vmatrix}$, then least integral value of $\frac{\Delta}{1000}$
 3. Let two parallel lines L_1 and L_2 with positive slope are tangent to the circle $C_1 : x^2 + y^2 - 2x - 16y + 64 = 0$. If L_1 is also tangent to the circle $C_2 : x^2 + y^2 - 2x + 2y - 2 = 0$ and equation of L_2 is $a\sqrt{a}x - by + c - a\sqrt{a} = 0$ where $a, b, c \in \mathbb{N}$ find value of $(a + b + c) - 10$
-

Space for Rough Work

4. Two sides of Rhombus OABC (lying entirely in the first quadrant or third quadrant) of area equal to 2sq. units are $x - \sqrt{3}y = 0$ and $\sqrt{3}x - y = 0$. Then sum of abscissa of possible co-ordinates of B is (O being origin)
5. If $4a^2 - 5b^2 + 6a + 1 = 0$ and the line $ax + by + c = 0$ touches a fixed circle then radius of circle is
6. Consider a fixed point A(5,2) and also B and C are two distinct variable points moving on $y - x = 0$ and $y = 0$ respectively. The points B and C are to be chosen such that $AB + BC + CA$ is minimum. If co-ordinates of B are (α, α) and $C(\beta, 0)$, then value of $\frac{\beta}{\alpha} =$
7. Let P be a point in the plane let $d_1(P)$ and $d_2(P)$ be the distance of the point P from lines $3x - 4y = 0$ and $4x + 3y = 0$ respectively. Area of region R consisting of all points P lying in first quadrant of the plane and satisfying $1 \leq d_1(P) + d_2(P) \leq 3$
8. If $\alpha + i\beta$ be the complex number satisfying $\left| \frac{z - 4i}{z - 2i} \right| = 1$ and $\left| \frac{z - 8 + 3i}{z + 3i} \right| = \frac{3}{5}$. Then sum of all possible value of α and β is

Space for Rough Work

Space for Rough Work

Space for Rough Work

NAME OF THE CANDIDATE

FORM NO.

I have read all the instructions
and shall abide by them._____
Signature of the CandidateI have verified the identity, name and Form
number of the candidate, and that question
paper and ORS codes are the same._____
Signature of the Invigilator

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